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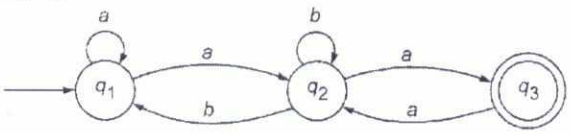
Thapar Institute of Engineering and Technology (TIET), Patiala

Department of Computer Science & Engineering (CSED)

SUMMER EST EXAMINATION

B. E. (Third Year)	Course Code: UCS701/UCT301 Course Name: TOC/FL & TA
20th July 2025	Friday, 09:00 AM-12:00 PM
Time: 03:00 Hrs., Max. Marks: 40	Name of Faculty: Dr. Shailendra Tiwari

Note: Attempt any EIGHT questions. Assume missing data, if any suitably.

1.	Reduce the following grammar to CNF: $S \rightarrow ASA \mid bA, \quad A \rightarrow B \mid S, \quad B \rightarrow c$	[5]
2.	Show that $L = \{a^i b^j \mid j = i^2\}$ is not context-free language.	[5]
3.	Construct a **PDA** accepting $\{a^n b^m a^n \mid m, n \geq 1\}$ by null store.	[5]
4.	Construct the corresponding context-free grammar of above given Q.3 that accepting the same set.	[5]
5.	Design a Turing machine for the languages $\{1^n 2^n 3^n \mid n \geq 1\}$.	[5]
6.	Given the following grammar $S \rightarrow a \mid YZ$ $Z \rightarrow ZY \mid a$ $Y \rightarrow b \mid ZZ \mid YY$ Apply CYK algorithm to check whether $w = babba$ belongs to this language or not?	[5]
7.	Consider the transition system given in Fig.1. Prove that the strings recognized are **(a + a(b + aa)*b)* a(b + aa)* a.**  Fig.1	[5]
8.	Consider the Mealy machine described by the transition table given by Table below. Construct a Moore machine which is equivalent to the Mealy machine.	[5]
9.	Design a DFA that accepts only those words that begin or end with a double letter.	[5]

Present state	*Next state*			
	Input a = 0		*Input a = 1*	
	state	*output*	*state*	*output*
$\rightarrow q_1$	q_3	0	q_2	0
q_2	q_1	1	q_4	0
q_3	q_2	1	q_1	1
q_4	q_4	1	q_3	0

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